Summary in Graph

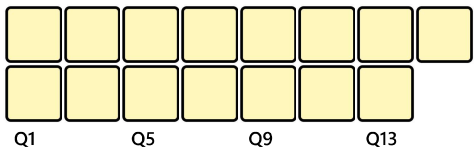
Exam Summary (GO Classes DA Test Series 2026 | Data Structures and Algorithms | Topic Wise Test 6)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (35) out of 350 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the given two statements.

- $S1$: Depth-first search is asymptotically faster than breadth-first search.
- $S2$: Deleting an element from a binary search tree takes $O(\lg n)$ time in the worst case.

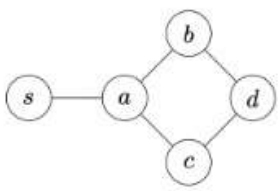
Which of the following is correct about $S1$ and $S2$?

A. $S1$ is correct but $S2$ is wrong.
B. $S2$ is correct but $S1$ is wrong.
C. Both are correct
D. Both are False

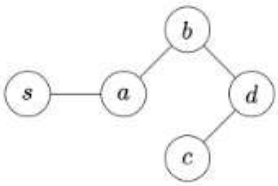
Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 02sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

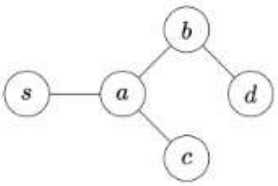
Consider the given graph G. Traversal trees $T1$ and $T2$ (given below) are made by DFS or BFS traversals starting from s..



Graph G



Traversal Tree T1



Traversal Tree T2

Which of the following is/are the correct option?

A. $T1$ is possible only using DFS on G
B. $T1$ is possible using DFS or BFS on G
C. $T2$ is possible only using BFS on G
D. $T2$ is possible using DFS or BFS on G

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Let $G = (V, E)$ be a simple undirected graph, and s be a particular vertex in it called the source. For $x \in V$, let $d(x)$ denote the shortest distance in G from s to x. A breadth-first search (BFS) is performed starting at s.

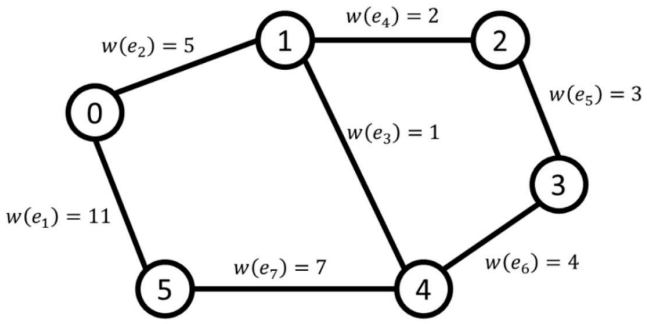
Which of the following is/are ALWAYS true?

A. There are no back edges
B. There are no forward edges
C. For each tree edge (u, v), we have $d[v] = d[u] + 1$
D. For each cross edge (u, v), we have $d[v] = d[u] + 1$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the below weighted graph where weight of the edge e is written as $w(e)$.



If we run Dijkstra's algorithm with $s = 0$, in which order will the vertices be deleted from the Priority Queue?

A. $0, 1, 2, 3, 4, 5$
B. $0, 1, 4, 2, 3, 5$
C. $0, 1, 5, 4, 2, 3$
D. $0, 1, 2, 4, 5, 3$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider performing a depth-first search (DFS) on an undirected and unweighted graph G starting at vertex s. For any vertex u in G, $d[u]$ is the length of the shortest path from s to u. Let (u, v) be an edge in G such that $d[u] < d[v]$. If the edge (u, v) is explored first in the direction from u to v during the above DFS, then (u, v) becomes a _____ edge.

A. tree
B. cross
C. back
D. gray

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following is/are TRUE ?

A. If we perform DFS on an undirected graph, there are no cross edges.
B. If the DFS tree has no back edges, then there are no cycles in the graph.
C. Dijkstra's algorithm will always work correctly on any graph that has at most two negative edges.
D. Dijkstra's algorithm will always work correctly on any graph with negative edges but no negative cycle.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Which of the following is correct option about $\mathrm{S1}$ and $\mathrm{S2}$?

- $\mathrm{S1}$: If G is a weighted graph with n vertices and m edges that does *not* contain negative weight cycle, then the iteration of the Bellman-Ford algorithm will reach a fixed point in at most $n - 1$ rounds.

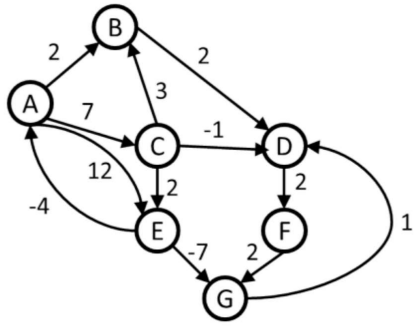
- $S2$: If G is a weighted graph with n vertices and m edges that *does* contain negative-weight cycle, then for *every* vertex v in G the shortest path from v to t in G containing n edges is strictly shorter than the shortest path from v to t in G containing $n-1$ edges.

A. $S1$ is TRUE but $S2$ is False
B. $S2$ is TRUE but $S1$ is False
C. Both are True
D. Both are False

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following directed, weighted graph:



Even though the graph has negative weight edges, we use Dijkstra's algorithm to calculate supposedly shortest paths from A to every other vertex.
Which of the following is/are correct?

A. The order in which vertices are added to the shortest path tree using Dijkstra's algorithm is : A B D F G C E.
B. The order in which vertices are added to the shortest path tree using Dijkstra's algorithm is : A B D F C G E.
C. Dijkstra's algorithm finds the wrong path to some of the vertices.
D. Dijkstra's algorithm finds the right path to all of the vertices.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following is/are TRUE?

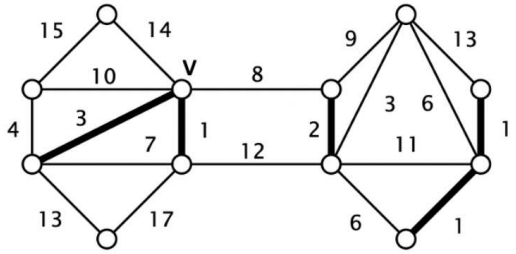
A. Dijkstra's algorithm may not terminate if the graph contains negative-weight edges.
B. Given a graph $G = (V, E)$ with positive edge weights, the Bellman-Ford algorithm and Dijkstra's algorithm can produce different shortest-path trees despite always producing the same shortest-path weights.
C. The Bellman-Ford algorithm applies to instances of the single-source shortest path problem which do not have a negative-weight directed cycle, but it does not detect the existence of a negative-weight directed cycle if there is one.
D. On a connected, directed graph with only positive edge weights, Bellman-Ford runs asymptotically as fast as Dijkstra.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

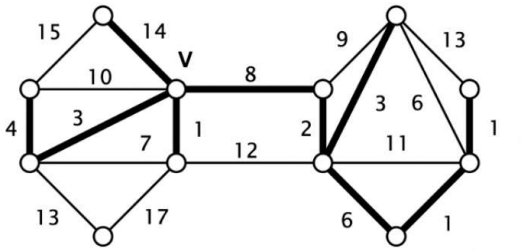
Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Each of the figures below represents a partial spanning tree with bold edges. Determine whether it could possibly be obtained from (a prematurely stopped) Prim's algorithm, (a prematurely stopped) Kruskal's algorithm, Both or Neither.
Prim's algorithm starts with vertex V in all cases.

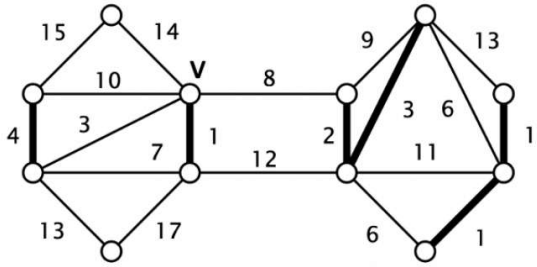
T1 :



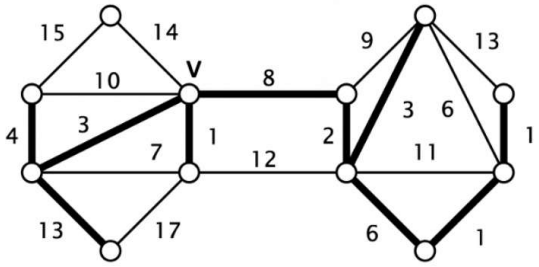
T2 :



T3 :



T4 :



A. T1 - Kruskal's only, T2- Prims only, T3- Neither, T4 - Both
B. T1 - Kruskal's only, T2- Prims only, T3 - Kruskal's only, T4 - Kruskal's only
C. T1 - Kruskal's only, T2- Neither, T3-Neither, T4 - Both
D. T1 - Kruskal's only, T2- Neither, T3- Kruskal's only, T4 - Prims only

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Graph Theory

For which of the following does there exist a simple undirected graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ satisfying the specified conditions?

I. G has 3 components, 20 vertices and 16 edges.
II. G has 12 vertices, 28 edges and the degree of each vertex is either 3 or 6.

A. Only I
B. Only II
C. Both
D. None

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Let $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ be a weighted directed graph. The shortest path from a node $s \in \mathrm{V}$ to a node $t \in \mathrm{V}$ will remain unchanged if: (Multiple options may be correct).
In all options $w(v, u)$ represents the edge weight of the edge $v \to u$ for $u, v \in \mathrm{V}$.

A. Each edge weight $w(v, u)$ is replaced by $\mathrm{C} \cdot w(v, u)$ for a constant $\mathrm{C} > 0$.
B. Each edge weight $w(v, u)$ is replaced by $w(v, u) + \mathrm{C}$ for a constant $\mathrm{C} > 0$.
C. Each edge weight $w(v, u)$ is replaced by $w(v, u) - \mathrm{C}$ for a constant $\mathrm{C} > 0$.
D. Each edge weight $w(v, u)$ is replaced by $w(v, u)/\mathrm{C}$ for a constant $\mathrm{C} > 0$.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

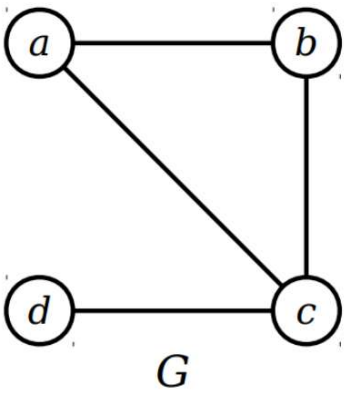
Which of the followings are CORRECT for Depth-First Search (DFS) on the graphs?
let n be greater than 2 in all options.

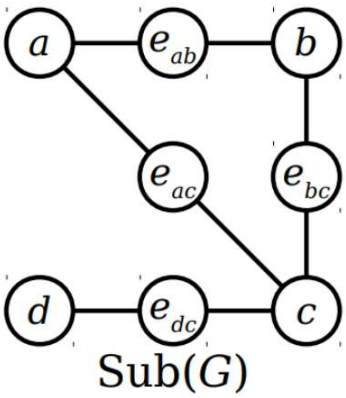
A. DFS on a directed graph with n vertices and n edges is guaranteed to find at least one back edge.
B. DFS on an undirected graph with n vertices and n edges is guaranteed to find at least one back edge.
C. DFS on an undirected graph with n vertices and n edges may find a cross edge.
D. DFS on a directed graph with n vertices and n edges may find one cross edge.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

A way to transform one undirected simple graph into another is by obtaining its subdivision. Intuitively, the subdivision of an undirected graph G, denoted $\mathrm{Sub}(\mathrm{G})$, is the graph formed by taking each edge in G, breaking the edge in half, and joining the two halves together by adding a node into the middle of the edge.
For example, here's a graph G and its subdivision:





Formally speaking, given a graph G, we can form its subdivision $\mathrm{Sub}(G)$ as follows. For each edge $\{u, v\} \in \mathrm{E}$, create a new node e_{uv}. Then, replace each edge $\{u, v\}$ with two edges $\{u, e_{uv}\}$ and $\{e_{uv}, v\}$. This resulting graph is $\mathrm{Sub}(\mathrm{G})$. (Intuitively, the subdivision of G is formed by "splitting" each edge into two edges, with a new node inserted in the middle.)

Which of the following is/are true about the subdivision of an undirected simple graph?

A. For every undirected simple graph G, the $\mathrm{Sub}(\mathrm{G})$ is always bipartite.
B. If G is a planar graph, then $\mathrm{Sub}(\mathrm{G})$ is also planar.
C. If G contains an even length cycle then $\mathrm{Sub}(\mathrm{G})$ also contains an even length cycle.
D. If $\mathrm{Sub}(\mathrm{G})$ contains an even length cycle then G also contains an even length cycle.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider a directed graph G with a source vertex s, a destination t, and nonnegative edge lengths. Under what conditions is the shortest $s - t$ path guaranteed to be unique?

A. When all edge lengths are distinct positive integers.
B. When all edges lengths are distinct positive integers and the graph G contains no directed cycles.
C. When all edge lengths are distinct powers of 2.
D. None of the other options are correct.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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